

Lab 4 (CS201)

1. **Limitations of the Euler's method** So far, the Euler method has been used to solve initial-value-problems numerically. In this lab session, some limitations of the method are illustrated by means of examples.

Airy Equation: Consider the equation,

$$\frac{d^2x}{dt^2} = tx(t).$$

- (a) Integrate the equation between $t = -30$ to $t = 2$ by Euler's method for number of subdivisions $n = 1000, 3000, 5000, 10000$. On the same graph(s), solve the initial value problem using scipy's inbuilt solve_ivp solver (try the Radau method) or Matlab's ode15s (or similar).
 - (b) Can you observe the difficulty in making the Euler method solution match closely with the more accurate solution obtained using in-built solvers? This problem illustrates the limitations of the Euler method and demonstrates why more sophisticated methods are desirable.
2. Consider the following equation,

$$\dot{x} = ax^2 - bx^3, \quad x(0) = \delta,$$

where δ is a small quantity.

- (a) Numerically integrate the above equation from $t = 0$ to $t = 1.5/\delta$ for $\delta = 0.1, 0.01, 0.001$ and for $a = b = 1$. You should be seeing a sharp increase at around $t \sim \delta^{-1}$ (i.e, this point changes with δ). Do you have to decrease the step size for each successive δ to obtain a smooth solution? If so, why?
- (b) Why is the sudden increase occurring at $t \sim \delta$?
(Warning: This is a slightly challenging question. Think about this exercise in the following way. You have done some numerical experiments/simulations on a system provided to you and have observed certain features. You are trying to explain whether there is any theoretical reason behind these observations.
Think about the logistic equation, $\dot{x} = ax - bx^2$, $x(0) = \delta$. Do you see the region where the “ramp”/increase occurs change with δ ?